Machine-Dependent Weighted Chairman Assignment: Counterexamples to Two Conjectures

Angus Muffatti

23 July 2026

Abstract

Liu and Reis conjectured that every fractional assignment with positive machine-dependent weights admits an integral assignment whose discrepancy on every row and every prefix is at most the largest weight. We give an exact rational counterexample with 11 rows and 15 columns. Every fractional column has support two, all weights belong to $\{1/1000, 1/2, 1\}$, and the largest weight is one. A three-column detector gadget forces one decision in each of five blocks. The five forced decisions make an accumulator row attain discrepancy at least $619/600 > 1$. The proof allows integral assignments outside the fractional support, and therefore refutes the unrestricted machine-dependent conjecture itself as well as the stronger unified support-preserving formulation.

1 Introduction

Let $x \in [0,1]^{m\times n}$ be a fractional assignment with

$$\sum_{i=1}^{m} x_{ij} = 1 \qquad (j = 1, \ldots, n).$$

An integral assignment $y \in \{0,1\}^{m\times n}$ satisfies the same column sums. Given positive machine-dependent weights d_{ij}, define the row-prefix discrepancy

$$\Delta_i(t) = \sum_{j=1}^{t} d_{ij}(x_{ij} - y_{ij}).$$

Liu and Reis proposed the following extension of the weighted chairman assignment problem [1].

Conjecture 1 (Liu–Reis, Conjecture 19)**.** For every fractional assignment x and every positive weight matrix d, there is an integral assignment y such that

$$|\Delta_i(t)| \le D := \max_{i,j} d_{ij} \qquad \text{for every row } i \text{ and prefix } t.$$

Their Conjecture 21 gives a unified statement with support preservation and more general prescribed integral column sums. (Conjecture numbers follow the ITCS 2026 proceedings version [1]; in the arXiv preprint 2511.18546 the same statements appear as Conjectures 3 and 5.) We show that already Conjecture 1 is false.

Theorem 2. *There is a rational instance with $m = 11$, $n = 15$, and $D = 1$ for which every integral assignment y has*

$$|\Delta_i(t)| > 1$$

for some row i and prefix t. Every fractional column has support two and every weight is strictly positive.

Since the theorem excludes all integral assignments, including those outside the support of x, it immediately implies the following.

Corollary 3. *The unified support-preserving statement of Liu and Reis (Conjecture 21) is false, already in the one-winner-per-column case.*

2 The instance

The rows are

$$B, A_1, H_1, A_2, H_2, \ldots, A_5, H_5,$$

and the columns occur in five consecutive blocks

$$J_k, K_k, L_k \qquad (k = 1, \ldots, 5).$$

Set

$$\eta = \frac{1}{1000}.$$

Every fractional entry not listed below is zero, and every weight not listed below equals η. Thus all weights are strictly positive.

For each block k, define the nonzero fractional entries and exceptional weights by Table 1.

Table 1: One block of the counterexample. Here $A = A_k$ and $H = H_k$.

column	nonzero fractional entries	exceptional weights
J_k	$x_{A,J_k} = 19/24$, $x_{B,J_k} = 5/24$	$d_{A,J_k} = d_{B,J_k} = 1$
K_k	$x_{A,K_k} = 1/2$, $x_{H,K_k} = 1/2$	$d_{A,K_k} = 1/2$, $d_{H,K_k} = 1$
L_k	$x_{A,L_k} = 23/48$, $x_{H,L_k} = 25/48$	$d_{A,L_k} = d_{H,L_k} = 1$

Every column sums to one, every column has support two, and

$$d_{ij} \in \left\{ \frac{1}{1000}, \frac{1}{2}, 1 \right\}, \qquad D = 1.$$

3 The forcing gadget

Fix a block k. Before column J_k, the fresh rows A_k and H_k have zero fractional entries in every preceding column. An integral assignment may nevertheless have assigned earlier columns to them. There are at most $3(k-1) \leq 12$ earlier columns, and each such off-support assignment contributes exactly $-\eta$. Hence immediately before J_k,

$$-12\eta \leq \Delta_{A_k} \leq 0, \qquad -12\eta \leq \Delta_{H_k} \leq 0. \tag{1}$$

Lemma 4. *Every integral assignment satisfying $|\Delta_i(t)| \leq 1$ on all rows and prefixes must assign J_k to A_k.*

Proof. Suppose that J_k is not assigned to A_k. Then A_k is unassigned on J_k and gains $19/24$. By (1), immediately after J_k,

$$\Delta_{A_k} \geq -12\eta + \frac{19}{24}. \tag{2}$$

The row H_k may itself receive J_k outside its support, which costs only η, so

$$\Delta_{H_k} \geq -13\eta. \tag{3}$$

Consider K_k. If it is not assigned to A_k, then A_k is unassigned and gains

$$d_{A_k,K_k} x_{A_k,K_k} = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

Together with (2), this gives

$$\Delta_{A_k} \geq -12\eta + \frac{19}{24} + \frac{1}{4} = \frac{25}{24} - \frac{12}{1000} > 1.$$

Thus a surviving assignment must send K_k to A_k.

After that forced choice,

$$\Delta_{A_k} \geq -12\eta + \frac{19}{24} - \frac{1}{4} = -12\eta + \frac{13}{24}, \tag{4}$$

$$\Delta_{H_k} \geq -13\eta + \frac{1}{2}. \tag{5}$$

It remains to assign L_k. If L_k is assigned to H_k, then A_k is unassigned and gains $23/48$. By (4),

$$\Delta_{A_k} \geq -12\eta + \frac{13}{24} + \frac{23}{48} = \frac{49}{48} - \frac{12}{1000} > 1.$$

If L_k is not assigned to H_k, then H_k is unassigned and gains $25/48$. By (5),

$$\Delta_{H_k} \geq -13\eta + \frac{1}{2} + \frac{25}{48} = \frac{49}{48} - \frac{13}{1000} = \frac{6047}{6000} > 1.$$

Every possible assignment of L_k therefore violates the unit bound. The assumption $J_k \not\to A_k$ was impossible. $\square$

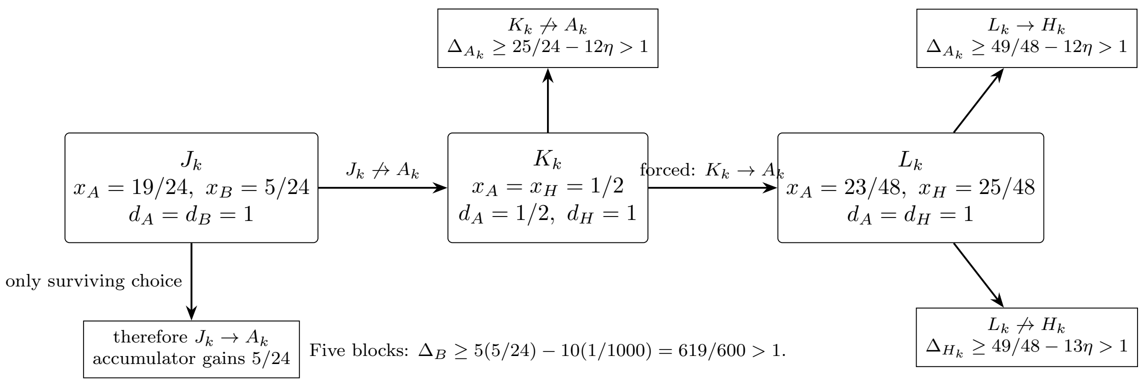

Figure 1: The three-column forcing gadget. The inequalities include the worst possible losses from assignments outside the fractional support.

4 Accumulator contradiction

Apply Lemma 4 independently to all five blocks. Every hypothetical discrepancy-good assignment must satisfy

$$J_k \longmapsto A_k \qquad (k = 1, \ldots, 5).$$

The accumulator row B is therefore unassigned on every J_k. Since $x_{B,J_k} = 5/24$ and $d_{B,J_k} = 1$, each block increases its discrepancy by $5/24$.

The only possible decreases occur when one of the ten detector columns K_k, L_k is assigned to B. On each such column $x_{B,j} = 0$ and $d_{B,j} = \eta$, so the decrease is at most η. At the final prefix,

$$\Delta_B(15) \geq 5 \cdot \frac{5}{24} - 10\eta = \frac{25}{24} - \frac{1}{100} = \frac{619}{600} > 1.$$

This contradiction proves Theorem 2.

5 A parametric family

The constants are instances of a simple open parameter region. In one block, choose

	x_A	x_H or x_B	d_A	d_H or d_B
J	$1-p$	p (B)	1	1
K	$1/2$	$1/2$ (H)	a	1
L	$1-q$	q (H)	1	1.

The detector works, before the small η corrections, whenever

$$\frac{a}{2} > p, \qquad \frac{1}{2} < q < 1 - p - \frac{a}{2}.$$

Indeed, the three forcing margins are

$$\frac{a}{2} - p, \qquad 1 - p - \frac{a}{2} - q, \qquad q - \frac{1}{2}.$$

With N blocks the accumulator margin is $Np - 1$, so we also require $Np > 1$. Once these inequalities are strict, $\eta > 0$ can be chosen small enough to absorb every off-support assignment.

Our rational choice is

$$N = 5, \qquad p = \frac{5}{24}, \qquad a = \frac{1}{2}, \qquad q = \frac{25}{48}, \qquad \eta = \frac{1}{1000}.$$

The ideal margins are respectively $1/24, 1/48, 1/48$, while the accumulator margin is $1/24$.

6 Discussion

The counterexample exploits exactly what changes when a common job weight is replaced by machine-dependent weights. With a common weight, assigning a job redistributes a conserved amount among rows. Here the K column has weight $1/2$ on A_k but weight 1 on H_k, permitting the detector to amplify a prefix imbalance. The small value η keeps every off-support weight positive without allowing unsupported choices to repair the forcing chain.

The result does not address the ordinary weighted-carpooling conjecture, in which a column has one common weight across all rows. It rules out the machine-dependent extension and every stronger statement containing it.

Reproducibility and disclosure

The accompanying repository contains the full 11×15 rational instance as JSON, an exact `Fraction`-based verifier for every inequality in the forcing certificate, and an independent mixed-integer infeasibility check. A generative AI system (GPT-5.6 Pro) assisted with exploration, checking, code drafting, and exposition. It is not an author. The author assumes responsibility for independent verification, priority, attribution, and the final submitted text.

References

[1] S. Liu and V. Reis. Weighted chairman assignment and flow-time scheduling. In *17th Innovations in Theoretical Computer Science Conference (ITCS 2026)*, volume 362 of *Leibniz International Proceedings in Informatics*, pages 98:1–98:15. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2026. doi: 10.4230/LIPIcs.ITCS.2026.98. Preprint arXiv:2511.18546.